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SMART INDIA HACKATHON 2025

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Low Cost Machinery for Picking of Cotton

- **Problem Statement ID – 25266**
- **Problem Statement Title -**

  Low Cost Machinery for Picking of Cotton

- **Theme- Agriculture, FoodTech & Rural Development**
- **PS Category-** Hardware
- **Team Name (Registered on portal)** Mindsmiths

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PROPOSED SOLUTION

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Problem at Hand

Labor Intensive: Manual cotton picking accounts for 30–40% of production costs and requires long hours of repetitive work.

Rising Labor Shortage: Rural labor availability is declining, making cotton harvesting expensive and unreliable.

Mechanical Limitations: Industrial cotton harvesters cost ₹4–5 lakh, are heavy, and cannot operate in small or uneven fields typical in India.

Quality Challenges: Manual picking often results in higher trash content (>15%), reducing lint quality and market value.

Our Solution

EcoPick Web Demo

Why We Stand Out

- **Affordable & Accessible**: Complete smart cotton picker under ₹10,000, ideal for small and marginal farmers.
- **AI + IoT Precision:** Picks only mature bolls, monitors crop & soil health, and provides real-time analytics.
- **Portable & Safe:** Lightweight, ergonomic, solar-hybrid powered, with protective features for operator safety.

Key Features

AI Picking — **Detects and harvests only mature cotton bolls.**

IoT Analytics — **Real-time soil, crop, and yield monitoring.**

Ergonomic Design — **Lightweight backpack suitable for all operators.**

Predictive Maintenance — **Vibration & current sensors for early fault alerts.**

Clean Harvesting — **Anti-static sack with multi-stage dust filter.**

Geo-Tagged Insights — **GPS mapping for yield and productivity tracking.**

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TECHNICAL APPROACH

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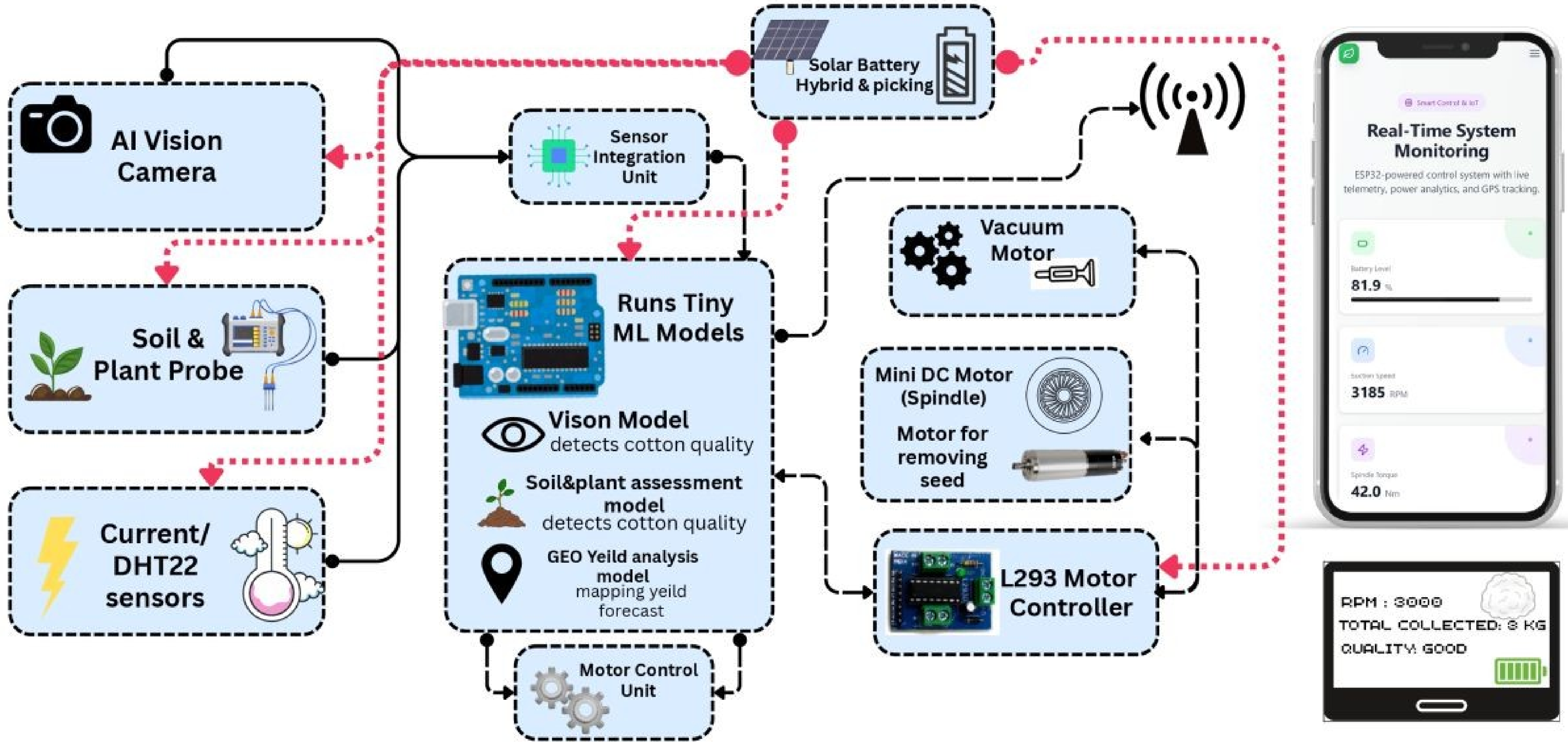

FEASIBILITY AND VIABILITY

Technical Feasibility

- **Built using commercially available components** (ESP32-C3, Li-ion battery, solar panel, vacuum motor, 3D-printed frame).
- AI insights & IoT-enabled Wi-Fi connectivity for real-time soil, pH, and yield data sync.
- **Low-power sensors** (NPK, moisture, pH) integrated without stressing battery life.

Economical Feasibility

Evaluation Parameter	Current Manual / Portable Picker	Proposed Smart System
Cost of Device	₹4,000 – ₹7,000	₹9,500 – ₹10,000
Manpower Required	2–3 workers	1 operator
Daily Picking Capacity	~17–25 kg	30–35 kg (selective, clean lint)
Trash Content	10–15 %	6–8 %
Labour Cost Reduction	—	≈ 60 %
Power Consumption	Battery only (limited)	Solar-battery hybrid (sustainable)

Maintenance & Safety Feasibility

- **Minimal maintenance –** few moving parts, easy to service with basic hand tools.
- **Electrostatic shielding**, dust filters, and insulated motor housing ensure user safety.
- **Predictive maintenance** via vibration and current sensing extends machine life.

Economic and Operational Viability

- **Total device cost under ₹10,000,** with daily operational cost < ₹40.
- **Payback within 1–2** harvests for 1–2 acre farms; ROI > 40%.
- **Lightweight backpack** design reduces operator fatigue by 30–40%.
- **Hip-mounted controller & multilingual mobile app.**

Implementation Viability

Factor	Support Element
Manufacturability	Simple low-cost machining or rural fab-lab assembly possible.
Component Availability	100 % components available in India.
Serviceability	Easy component replacement; no dependency on sophisticated tools.
Scalability	Modular add-ons (multi-nozzle picker, fiber sensor integration).
Deployment Readiness	Prototype ready for field testing; estimated commercial scaling possible within 12 months.

Environmental Viability

- **Low-heat, anti-static** materials meet agricultural safety standards.
- **Promotes climate-resilient** and low-carbon farming practices.
- **Supports Atmanirbhar Bharat & Digital India missions** through digital literacy and rural innovation.

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RESEARCH AND REFERENCES

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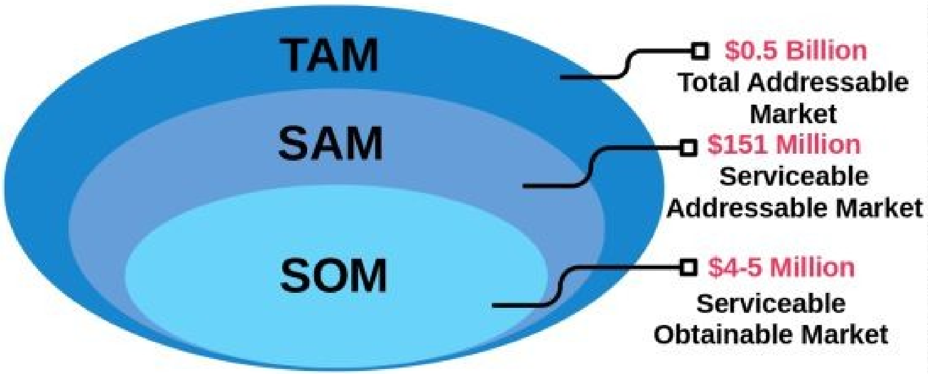

TAM — Total Addressable Market (Indian Small Farm Mechanization Market)

Under the assumption of 100% adoption,

we can calculate TAM as being : 1x0.5 Billion = $0.5 Billion

SAM — Serviceable Available Market (Indian Small Farm Mechanization of Cotton Farms)

Under the assumption of 100% adoption,

we can calculate SAM as being : 1.26x 120 = $151 Million

SOM — Serviceable Obtainable Market (1-year, 100% adoption)

we can calculate SAM as being : 1.26 m × 30% = ~378,000 farmers reachable. Then 10% conversion → ~37,800 units.

REFERENCE

1) J. M. Bennett, N. P. Woodhouse, T. Keller, T. A. Jensen and D. L. Antille, “Advances in cotton harvesting technology: a review and implications for the John Deere round baler cotton picker,” Journal of Cotton Science, vol. 19, no. 2, pp. 225-249, 2015.

2) J. Bankar, G. Nimbalkar, M. Nimbalkar, A. Hon, S. Ghodke and S. Tamboli, “Design & Development of Smart Cotton-Picking Robot Using ML & Suction Mechanism,” 2025 SAE Media Group TechBriefs Contest – Robotics & Automation, June 1 2025.

3) C. Zhang et al., “The Development of Wear Characteristics of the Picking Hook Teeth,” Wear, vol. XXXXX, 2024. doi:10.1016/j.wear.2024.00060.

4) A. D. Abdazimov, S. S. Radjabov and N. N. Omonov, “Automation of agrotechnical assessment of cotton harvesting machines,” Journal of Physics: Conference Series, vol. 1260, no. 3, p. 032001, 2019. doi:10.1088/1742-6596/1260/3/032001.

LINKS

1) Documentation

2) TAM SAM SOM Documentation

3) Blender Model Images

4) Budget Report

5) Demo Video Link - Web Demo

6) Actual Website